Comprehensive Analysis of the Dual-Slope Radio Propagation Model in Vehicular Communications (V2X): Theory, Simulator Implementation, and 3GPP TR 38.901 Conformance Audit

RustV2XSimulator Engineering & Academic Research Documentation
Autonomous Systems & Vehicular Networks Simulation Framework

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Abstract

Radio wave propagation in vehicular ad-hoc networks (VANETs) and Cooperative Intelligent Transport Systems (C-ITS) is dominated by ground-plane asphalt reflections, low antenna elevations, and severe urban obstacle shadowing. In this treatise, we provide an exhaustive physical and mathematical analysis of the Dual-Slope Log-Distance Path Loss model. We first derive the model from Maxwellian boundary conditions and the asymptotic Two-Ray Ground Reflection formulation, proving why received power transitions from an inverse-square distance decay (d^{-2}) to an inverse-quartic decay (d^{-4}) at the Fresnel zone clearance boundary. We then document the native Rust implementation within `rustv2xsimulator`, detailing its integration with Rayon-parallelized link evaluation, building wall penetration ray-casting, and ITU-R P.526 knife-edge corner diffraction. Finally, we perform a strict standards conformance audit against 3GPP TR 38.901 (ETSI TR 138 901 v18.1.0). We report and explain eight critical incongruencies between the simulator's current physical layer parameters (tuned for strict ns-3 v3.42 and IEEE 802.11p empirical parity) and the 3GPP cellular infrastructure standard, providing an architectural blueprint for native multi-standard support.

1 Introduction

Vehicular communications (V2X) operating in the 5.9 GHz Intelligent Transport Systems (ITS) band must accurately model electromagnetic wave attenuation over roadway geometries. Unlike cellular downlink and uplink channels where base stations (BS) are elevated at 10 m to 35 m on masts or building rooftops, peer-to-peer Vehicle-to-Vehicle (V2V) links operate with both transmitter and receiver antennas positioned at near-ground heights ($h_{tx}, h_{rx} \in [1.2, 1.8]$ m).

The presence of the conductive ground plane creates a dominant reflected ray that interferes destructively with the direct line-of-sight (LOS) wave once a characteristic breakpoint distance d_{break} is reached. This phenomenon produces a distinct piecewise attenuation curve characterized by two logarithmic slopes.

In this document, we comprehensively examine:

1. The electromagnetic derivation and physics of the dual-slope phenomenon.
2. The analytical formulation and log-normal shadow fading modeling.
3. The native Rust implementation within `rustv2xsimulator`.
4. A line-by-line conformance comparison against the 3GPP TR 38.901 / ETSI TR 138 901 v18.1.0 specification.

5. An engineering incongruence report highlighting why divergences exist and how they relate to ns-3 v3.42 parity.

2 Electromagnetic and Physical Foundations

2.1 Two-Ray Ground Reflection Geometry

Consider a transmitter at Cartesian coordinate $(0, 0, h_{tx})$ and a receiver at $(d, 0, h_{rx})$ separated by ground planar distance d. The direct ray distance d_{dir} and ground-reflected ray distance d_{ref} are:

$$d_{\text{dir}} = \sqrt{d^2 + (h_{tx} - h_{rx})^2} \approx d + \frac{(h_{tx} - h_{rx})^2}{2d} \tag{1}$$

$$d_{\text{ref}} = \sqrt{d^2 + (h_{tx} + h_{rx})^2} \approx d + \frac{(h_{tx} + h_{rx})^2}{2d} \tag{2}$$

valid under the paraxial approximation $d \gg h_{tx} + h_{rx}$.

The differential path length Δd is:

$$\Delta d = d_{\text{ref}} - d_{\text{dir}} \approx \frac{(h_{tx} + h_{rx})^2 - (h_{tx} - h_{rx})^2}{2d} = \frac{2h_{tx}h_{rx}}{d} \tag{3}$$

For a carrier frequency f_c with wavelength $\lambda = c/f_c$, the phase differential $\Delta\phi$ between the two rays is:

$$\Delta\phi = \frac{2\pi}{\lambda}\Delta d \approx \frac{4\pi h_{tx}h_{rx}}{\lambda d} \tag{4}$$

2.2 Total Received Field and Asymptotic Regimes

Assuming horizontal polarization or grazing incidence where the Fresnel reflection coefficient $\Gamma \approx -1$, the superposition of direct and reflected electric fields yields:

$$E_{\text{total}} \approx \frac{E_0}{d}\left(1 - e^{-j\Delta\phi}\right) = \frac{2E_0}{d}\sin\left(\frac{\Delta\phi}{2}\right)e^{-j\left(\frac{\Delta\phi - \pi}{2}\right)} \tag{5}$$

The field magnitude is modulated by $\sin\left(\frac{2\pi h_{tx}h_{rx}}{\lambda d}\right)$:

1. **Near-Field Regime ($d < d_{\text{break}}$):** Here, $\Delta\phi > \pi$. The sine argument changes rapidly with distance, creating alternating constructive and destructive interference fringes (Fresnel oscillations). Averaged over local spatial windows, the median received power attenuates in accordance with the Friis free-space spreading law:

$$P_{rx}(d) \propto d^{-\gamma_1}, \quad \gamma_1 \approx 2.0 \tag{6}$$

2. **Far-Field Regime ($d > d_{\text{break}}$):** Here, $\Delta\phi \ll 1$. Applying the small-angle expansion $\sin(\theta) \approx \theta$:

$$|E_{\text{total}}| \approx \frac{2E_0}{d}\left(\frac{2\pi h_{tx}h_{rx}}{\lambda d}\right) = \frac{4\pi E_0 h_{tx}h_{rx}}{\lambda d^2} \tag{7}$$

Since received power $P_{rx} \propto |E_{\text{total}}|^2$:

$$P_{rx}(d) \propto d^{-4} \implies P_{rx}(d) \propto d^{-\gamma_2}, \quad \gamma_2 \approx 4.0 \tag{8}$$

2.3 Breakpoint Distance Derivation

The transition between oscillatory free-space spreading and the d^{-4} ground cancellation regime occurs at phase boundary $\Delta\phi = \pi$ (or path difference $\Delta d = \lambda/2$):

$$\Delta\phi = \frac{4\pi h_{tx} h_{rx}}{\lambda d_{\text{break}}} = \pi \implies d_{\text{break}} = \frac{4h_{tx}h_{rx}}{\lambda} = \frac{4h_{tx}h_{rx}f_c}{c} \tag{9}$$

For standard vehicular rooftop antennas ($h_{tx} = h_{rx} = 1.5\,\text{m}$) at $f_c = 5.9\,\text{GHz}$ ($\lambda \approx 0.0508\,\text{m}$):

$$d_{\text{break}} = \frac{4 \cdot 1.5 \cdot 1.5}{0.0508} \approx 177.17\,\text{m} \tag{10}$$

In practical vehicular environments, surface roughness, road curvature, and surrounding traffic clutter perturb the ideal ground reflection, causing the effective empirical breakpoint to sit between 80 m and 150 m.

3 Mathematical Formulation

3.1 Deterministic Dual-Slope Path Loss

Let $d_0 = 1.0\,\text{m}$ be the reference distance. The free-space path loss at d_0 is:

$$PL_0 = 20\log_{10}\left(\frac{4\pi d_0 f_c}{c}\right) \tag{11}$$

At 5.9 GHz, $PL_0 \approx 47.854\,\text{dB}$.

The deterministic path loss $PL_{\text{det}}(d)$ for $d \geq d_0$ is defined piecewise:

$$PL_{\text{det}}(d) = \begin{cases} PL_0 + 10\gamma_1 \log_{10}(d), & d_0 \leq d \leq d_{\text{break}} \\ PL_{\text{break}} + 10\gamma_2 \log_{10}\left(\frac{d}{d_{\text{break}}}\right), & d > d_{\text{break}} \end{cases} \tag{12}$$

where:

$$PL_{\text{break}} = PL_0 + 10\gamma_1 \log_{10}(d_{\text{break}}) \tag{13}$$

By construction:

$$\lim_{d \to d_{\text{break}}^-} PL_{\text{det}}(d) = \lim_{d \to d_{\text{break}}^+} PL_{\text{det}}(d) = PL_{\text{break}} \tag{14}$$

ensuring exact C^0 continuity across the breakpoint boundary.

3.2 Large-Scale Shadow Fading

To represent shadowing from foliage, topography, and roadway clutter:

$$PL(d) = \max\left(0.0,\, PL_{\text{det}}(d) + X_\sigma\right), \quad X_\sigma \sim \mathcal{N}(0, \sigma^2) \tag{15}$$

where $\sigma = \sigma_{\text{LoS}}$ under Line-of-Sight conditions, and $\sigma = \sigma_{\text{NLoS}}$ under Non-Line-of-Sight conditions.

4 Implementation in RustV2XSimulator

4.1 Physical Layer Data Structure

Implemented in `src/phy/path_loss.rs`:

```
pub struct DualSlopePathLoss {
    pub carrier_frequency_hz: f64,
    pub d_break: f64,
    pub gamma1: f64,
    pub gamma2: f64,
    pub sigma_los: f64,
    pub sigma_nlos: f64,
    pub pl_0: f64,
}
```

The default constructor instantiates:

- $f_c = 5.9 \times 10^9$ Hz
- $d_{\text{break}} = 120.0$ m
- $\gamma_1 = 2.05$, $\gamma_2 = 3.90$
- $\sigma_{\text{LoS}} = 3.1$ dB, $\sigma_{\text{NLoS}} = 8.3$ dB

4.2 End-to-End Channel Pipeline

Within `RadioMedium::evaluate_transmission`, the engine evaluates:

1. **Ray Casting:** Intersects transmitter-receiver vector with OpenStreetMap 2D building polygons, calculating concrete penetration distance d_{pen}.
2. **Dual-Slope Loss:** Evaluates $PL_{\text{det}}(d)$.
3. **Shadow Fading:** Samples $X_\sigma \sim \mathcal{N}(0, \sigma^2)$ using deterministic PRNG hash $sim_seed \oplus (packet_id \ll 32 \mid rx_id) \oplus idx$.
4. **Wall Penetration:** Adds $PL_{\text{pen}} = \min(d_{\text{pen}} \times 0.85\,\text{dB/m}, 45.0\,\text{dB})$.
5. **Corner Diffraction:** Computes ITU-R P.526 knife-edge attenuation $J(\nu)$ around the nearest occluding vertex.
6. **Received Power:**

$$P_{rx} = P_{tx} - (PL_{\text{det}}(d) + X_\sigma + PL_{\text{pen}} + J(\nu)) \tag{16}$$

5 3GPP TR 38.901 (ETSI TR 138 901) Specification

5.1 Overview and Cellular Scope

3GPP TR 38.901 defines propagation channels for 5G NR across 0.5 GHz to 100 GHz. The document standardizes models for:

- Urban Micro (UMi) - Street Canyon
- Urban Macro (UMa)
- Rural Macro (RMa)
- Indoor Office (InH) and Indoor Factory (InF)

Crucially, Clause 6.4 notes: *"Develop a methodology considering that model extensions to D2D and V2V may be developed in future SI."* TR 38.901 focuses primarily on Base Station to User Terminal (BS-UT) infrastructure links.

Table 1: Standards Conformance Audit: RustV2XSimulator vs. 3GPP TR 38.901 (ETSI TR 138 901 v18.1.0)

Parameter / Dimension	**RustV2XSimulator**	**3GPP TR 38.901 (UMi LOS)**	**Compliance Assessment**
Link Topology	Peer-to-Peer V2V Sidelink	Cellular BS $\leftrightarrow$ UT Infrastructure	Divergent (V2V vs. BS-UT)
Breakpoint Distance (d_{break})	Static constant: 120.0 m	Dynamic: $4(h_{BS}-h_E)(h_{UT}-h_E)f_c/c$	Non-compliant (Static vs. Dynamic)
Near-Field Slope (γ_1)	$\gamma_1 = 2.05$	$\gamma_1 = 2.10$ ($10 \times 2.1 \log_{10} d$)	Incongruent ($\Delta = -0.05$)
Far-Field Slope (γ_2)	$\gamma_2 = 3.90$	$\gamma_2 = 4.00$ ($40 \log_{10} d$)	Incongruent ($\Delta = -0.10$)
Reference Offset PL_0 at 1 m	$20 \log_{10}(4\pi f_c/c) = 47.854$ dB	$32.4 + 20 \log_{10}(f_c) = 47.820$ dB	Minor Offset (0.034 dB)
Far-Field Continuity Offset	Relative division: $+10\gamma_2 \log_{10}(d/d_{\text{break}})$	Geometric offset: $-9.5 \log_{10}((d'_{BP})^2 + \Delta h^2)$	Incongruent Formulation
LOS Shadow Fading (σ_{LoS})	$\sigma_{\text{LoS}} = 3.1$ dB	$\sigma_{SF} = 4.0$ dB	Incongruent ($\Delta = -0.9$ dB)
NLOS Shadow Fading (σ_{NLoS})	$\sigma_{\text{NLoS}} = 8.3$ dB	$\sigma_{SF} = 7.82$ dB	Incongruent ($\Delta = +0.48$ dB)
Distance Coordinate	2D Planar $d_{2D} = \sqrt{\Delta x^2 + \Delta y^2}$	3D Slant $d_{3D} = \sqrt{d_{2D}^2 + \Delta h^2}$	Incongruent (2D vs. 3D)
NLOS Attenuation	GIS ray-casting + Knife-Edge Diff.	Empirical formula: $\max(PL_1, PL'_{NLOS})$	Architectural Divergence

5.2 Table 7.4.1-1: UMi - Street Canyon Formulations

For UMi LOS links, Table 7.4.1-1 specifies:

$$d'_{BP} = 4h'_{BS}h'_{UT}\frac{f_c}{c} \tag{17}$$

with effective antenna heights $h'_{BS} = h_{BS} - h_E$, $h'_{UT} = h_{UT} - h_E$, and $h_E = 1.0$ m.

The path loss equations (f_c in GHz, d in meters) are:

$$PL_1(d_{3D}) = 32.4 + 21.0 \log_{10}(d_{3D}) + 20 \log_{10}(f_c), \quad 10\,\text{m} \le d_{2D} \le d'_{BP} \tag{18}$$

$$\begin{aligned} PL_2(d_{3D}) = {} & 32.4 + 40.0 \log_{10}(d_{3D}) + 20 \log_{10}(f_c) \\ & - 9.5 \log_{10}\left((d'_{BP})^2 + (h_{BS} - h_{UT})^2\right), \quad d'_{BP} \le d_{2D} \le 5\,\text{km} \end{aligned} \tag{19}$$

with shadow fading standard deviation $\sigma_{SF} = 4.0$ dB.

For UMi NLOS links:

$$PL'_{NLOS}(d_{3D}) = 35.3 + 22.4 \log_{10}(d_{3D}) + 21.3 \log_{10}(f_c) - 0.3(h_{UT} - 1.5) \tag{20}$$

$$PL_{NLOS}(d_{3D}) = \max\left(PL_1(d_{3D}),\, PL'_{NLOS}(d_{3D})\right) \tag{21}$$

with $\sigma_{SF} = 7.82$ dB.

6 Standards Conformance Audit & Incongruence Report

Table 1 provides a comprehensive point-by-point comparison between the simulator implementation and 3GPP TR 38.901 Table 7.4.1-1.

6.1 Detailed Incongruence Analysis

1. **Topology Mismatch:** TR 38.901 is an infrastructure cellular model. V2V sidelink communications are specified in 3GPP TR 36.885 and TR 37.885.

2. **Static vs. Dynamic Breakpoint:** `rustv2xsimulator` uses a constant $d_{\text{break}} = 120.0\,\text{m}$. TR 38.901 scales d'_{BP} dynamically with effective antenna heights and frequency.

3. **Slope Exponents:** The simulator uses empirical 802.11p fits ($\gamma_1 = 2.05, \gamma_2 = 3.90$) instead of TR 38.901 values ($\gamma_1 = 2.10, \gamma_2 = 4.00$).

4. **Shadow Fading Variances:** The simulator uses empirical measurements ($\sigma_{\text{LoS}} = 3.1\,\text{dB}, \sigma_{\text{NLoS}} = 8.3\,\text{dB}$) rather than TR 38.901 values ($\sigma_{\text{LoS}} = 4.0\,\text{dB}, \sigma_{\text{NLoS}} = 7.82\,\text{dB}$).

5. **NLOS Modeling Philosophy:** The simulator computes geometric obstacle penetration ($d_{\text{pen}} \times 0.85\,\text{dB/m}$) and knife-edge diffraction from OpenStreetMap building polygons, whereas TR 38.901 uses empirical parametric macro-formulas.

7 Engineering Rationale: ns-3 Gold Standard Parity

The parameter choices in `rustv2xsimulator` were selected to achieve strict mathematical parity with the **ns-3 v3.42 gold standard** (`ns3::BreakpointPathlossModel`), validated in `tests/ns3_phy_parity_test.rs`. Across 80 distance bins up to 800 m, the simulator achieves a Packet Delivery Ratio Mean Absolute Error of MAE ≤ 0.021. Modifying these defaults to TR 38.901 without a configurable toggle would invalidate existing IEEE 802.11p regression baselines.

8 Proposed Architecture for Native TR 38.901 Support

We propose extending `src/phy/path_loss.rs` with an enumeration:

```
pub enum DualSlopeMode {
    Ns3VeinsEmpirical {
        d_break: f64,
        gamma1: f64,
        gamma2: f64,
        sigma_los: f64,
        sigma_nlos: f64,
    },
    Tr38901UmiStreetCanyon {
        h_bs: f64,
        h_ut: f64,
        h_e: f64,
    },
    Tr37885V2vSidelink {
        h_tx: f64,
        h_rx: f64,
    },
}
```

This maintains backward parity with ns-3 v3.42 while offering strict standards compliance for 3GPP cellular and sidelink evaluations.

9 Conclusion

The Dual-Slope Log-Distance model is essential for capturing two-ray ground reflection in vehicular communications. While the current implementation in `rustv2xsimulator` is mathematically sound, continuous, and verified against ns-3 v3.42, it exhibits eight specific incongruencies with 3GPP TR 38.901 Table 7.4.1-1 due to differing target topologies (peer V2V vs. cellular infrastructure). Implementing a dual-mode standard configuration will enable researchers to toggle seamlessly between empirical VANET baselines and 3GPP 5G NR standards.

References

[1] 3GPP, "Study on channel model for frequencies from 0.5 to 100 GHz," 3GPP TR 38.901 (ETSI TR 138 901) v18.1.0, Release 18, 2024.

[2] 3GPP, "Study on evaluation methodology of new V2X use cases for LTE and NR," 3GPP TR 37.885 v15.3.0, 2019.

[3] C. Sommer and F. Dressler, "Using the right two-ray model? A look at obstacle modeling in VANETs," in *Proc. ACM VANET*, 2011.

[4] T. Abbas, K. Sjöberg, J. Karedal, and F. Tufvesson, "Measurement based shadow fading model for vehicle-to-vehicle network simulations," *Int. J. Antennas Propag.*, 2012.